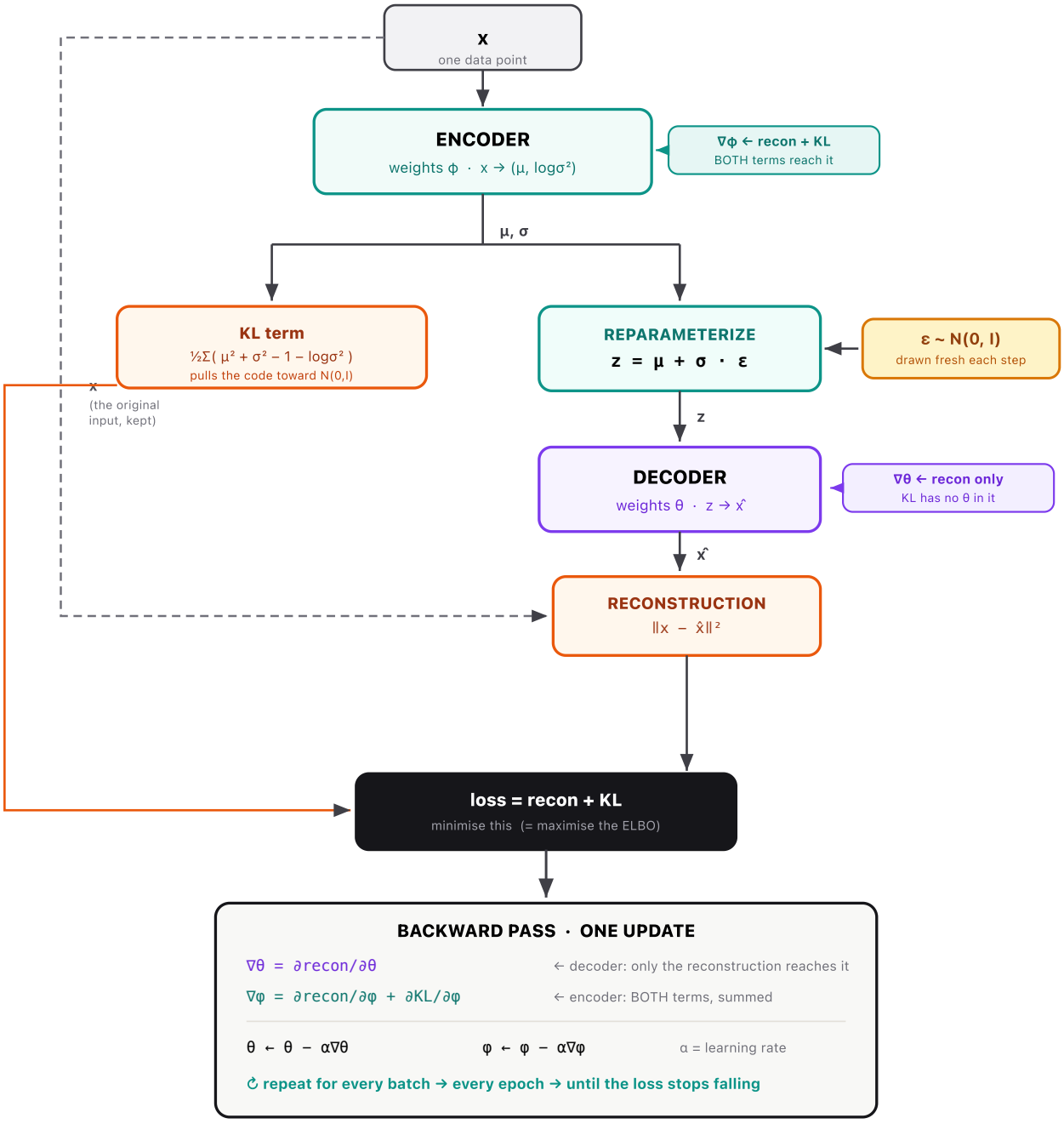
# How VAE training works — end to end

One pass = one forward sweep, one loss, one backward sweep, one update. Then repeat. **(page 1 of 2)**




the ∇ϕ path runs **backwards through z** — that is exactly why the reparameterization block exists. Without it this arrow is severed.

forward flow is **top → bottom**; gradients flow **bottom → up**, taking **different routes** for θ and for ϕ.

**Reading it:** the data **x** goes into the encoder, which emits the *parameters* of a Gaussian over the code (μ and logσ²) — not a code. A fresh noise sample **ε** is combined with them to produce **z**. The decoder maps **z** to a reconstruction **x̂**. The two loss terms are computed from different inputs (**x** & **x̂** for reconstruction; **μ** & **σ** for KL) and summed. One backward pass then routes **∇θ** into the decoder alone and **∇ϕ** into the encoder from both terms.